

Project Report

on

ShopNova: E-Commerce Web App

Front End Technologies (25CAH-655)

Submitted by

Hritik Raj
25MCH10279

Under the guidance of

Ms. Saranjeet Kaur

in partial fulfilment for the award of the degree of

**MASTER OF COMPUTER APPLICATIONS
ARTIFICIAL INTELLIGENCE & MACHINE
LEARNING**



Chandigarh University

April 2026

Certificate

This is to certify that Hritik Raj, a student of Master of Computer Applications (MCA) – Artificial Intelligence & Machine Learning, has successfully completed the Minor Project titled “ShopNova: E-Commerce Web App” under the esteemed guidance of Ms. Saranjeet Kaur, Assistant Professor, University Institute of Computing (UIC), Chandigarh University.

This project was undertaken as part of the academic curriculum and is submitted in partial fulfilment of the requirements for the MCA program. The work presented in this project is the result of independent development, programming practice, and dedication, demonstrating the student’s ability to apply theoretical knowledge to practical web application development.

The project, “**ShopNova: E-Commerce Web App**”, is a frontend-based web application developed using **HTML, CSS, and JavaScript**. It provides a dynamic online shopping experience with features such as product browsing, category filtering, search functionality, shopping cart management, wishlist system, and user authentication using local storage. The application effectively demonstrates modern UI/UX design, responsive layout techniques, state management, and event-driven programming.

I hereby confirm that this project is an original work carried out by the student and has not been submitted elsewhere for the award of any other degree, diploma, or certification.

Project Guide:

Ms. Saranjeet Kaur

Assistant Professor

University Institute of Computing
Chandigarh University

Acknowledgement

I would like to express my sincere gratitude to Chandigarh University and the University Institute of Computing (UIC) for providing me with the opportunity to undertake this project titled “ShopNova: E-Commerce Web App.”

I extend my heartfelt appreciation to my esteemed mentor, Ms. Saranjeet Kaur, Assistant Professor, for her invaluable guidance, continuous support, and insightful feedback throughout the development of this project. Her expertise in web technologies and constant motivation played a vital role in the successful completion of this work.

I am also deeply thankful to my friends and peers for their encouragement, constructive suggestions, and collaboration, which helped enhance the quality of this project. Finally, I would like to express my profound gratitude to my family for their unwavering support, patience, and motivation throughout this learning journey.

This project has been a valuable learning experience, helping me strengthen my understanding of frontend web development using HTML, CSS, and JavaScript, along with concepts such as responsive design, user interface development, and client-side state management. I hope this project serves as a foundation for further exploration and innovation in web application development.

Hritik Raj

MCA – Artificial Intelligence & Machine Learning
Chandigarh University

Contents

Section	Page Number
1. Introduction	1
2. Objective	2
3. Tools and Libraries Used	3
4. Implementation Steps	4 - 5
5. Screenshots of Code and Results	6 – 11
6. Conclusion	12
References	13

1. Introduction

In today's digital era, online shopping platforms have become an integral part of everyday life, providing users with convenience, accessibility, and a wide range of product choices. With the rapid growth of internet technologies, there is an increasing demand for efficient and user-friendly e-commerce applications that enhance the overall shopping experience. ShopNova: E-Commerce Web App is developed to address this need by offering a dynamic and interactive platform for online shopping.

This project focuses on designing and developing a frontend-based e-commerce web application using HTML, CSS, and JavaScript. The application enables users to browse products across multiple categories, search for specific items, and apply filters based on price, brand, and ratings. It also provides essential features such as a shopping cart system, wishlist management, and user authentication using local storage, allowing users to simulate a real-world online shopping environment.

The interface is designed with modern UI/UX principles to ensure smooth navigation and an engaging user experience. Responsive design techniques are implemented to make the application accessible across different devices. The project emphasizes event-driven programming and efficient state management to handle user interactions effectively.

Overall, ShopNova demonstrates how web technologies can be utilized to build a functional and visually appealing e-commerce platform. It serves as a practical implementation of frontend development concepts while showcasing the importance of usability, performance, and interactivity in modern web applications.

2. Objective

➤ To Develop an Interactive E-Commerce Platform

The primary objective of this project is to design and develop an interactive web-based Application that allows users to browse, search, and explore products efficiently in a Simulated online shopping environment.

➤ To Enhance User Experience through a Responsive UI

The project aims to create a visually appealing and user-friendly interface using **HTML, CSS, and JavaScript**, ensuring smooth navigation, responsiveness across devices, and an engaging shopping experience.

➤ To Implement Product Filtering and Sorting Mechanisms

The system incorporates features such as category-based filtering, price range selection, brand filtering, and sorting options (price, rating, newest) to help users find products quickly and efficiently.

➤ To Build Shopping Cart and Wishlist Functionality

The application provides essential e-commerce features like adding/removing items from the cart, managing quantities, saving favorite products in a wishlist, and dynamically updating the total cost.

➤ To Implement Client-Side State Management using Local Storage

The project uses browser local storage to store user data such as cart items, wishlist, and login details, ensuring data persistence even after page refresh.

➤ To Apply Frontend Development Concepts in a Real-World Scenario

This project demonstrates the practical application of frontend technologies, including event-driven programming, DOM manipulation, responsive design, and state management, to build a functional and scalable web application.

3. Tools and Libraries Used

➤ **HTML (HyperText Markup Language) :**

Used to create the basic structure of the web application, including layout elements such as navbar, product sections, forms, and pages.

➤ **CSS (Cascading Style Sheets) :**

Used for designing and styling the user interface, including colors, layouts, animations, responsiveness, and overall visual appearance.

➤ **JavaScript (Vanilla JavaScript) :**

Used to implement the core functionality of the application, including dynamic content rendering, event handling, filtering, sorting, and user interactions.

➤ **DOM Manipulation :**

JavaScript is used to dynamically update the Document Object Model (DOM), allowing real-time updates of products, cart items, wish list, and UI components.

➤ **Local Storage (Browser Storage) :**

Used to store user data such as cart items, wishlist, and login information, ensuring data persistence even after page refresh.

➤ **Event-Driven Programming :**

Implemented to handle user actions such as clicking buttons, searching products, applying filters, and managing cart operations.

➤ **Responsive Web Design :**

CSS media queries are used to ensure that the application works smoothly across different devices like desktops, tablets, and mobile phones.

➤ **UI/UX Design Principles :**

Applied to create an intuitive and user-friendly interface with smooth navigation, interactive elements, and enhanced user experience.

➤ **Modular JavaScript Functions :**

The project uses structured and reusable functions to maintain clean, organized, and scalable code.

4. Implementation Steps

➤ Project Planning and Setup

The project began with defining the main objective — to design a user-friendly and interactive e-commerce web application that simulates an online shopping experience. The development environment was set up using **HTML, CSS, and JavaScript**, focusing on building a frontend-only application without backend integration.

➤ Designing the User Interface (UI)

A modern and responsive interface was designed using CSS, including components such as the navigation bar, product grid, category filters, search bar, cart drawer, and modals for login and signup. Emphasis was placed on clean layout, visual appeal, and ease of navigation to enhance user experience.

➤ Creating the Application Structure

The project follows a modular structure using JavaScript functions to handle different operations such as rendering products, managing cart items, handling wishlist, applying filters, and updating the UI. Variables like `cartItems`, `wishlist`, `currentUser`, and `currentFilters` are used to manage the application state efficiently.

➤ Implementing Core Functionalities

Core features such as product listing, category filtering, search functionality, sorting options, cart management, and wishlist system were implemented using JavaScript. Event-driven programming is used to handle user actions like clicking buttons, selecting categories, and adding products to the cart.

➤ State Management using Local Storage

Browser local storage is used to store user data such as cart items, wishlist, and login details. This ensures that user data persists even after refreshing the page, providing a continuous and realistic shopping experience.

➤ Dynamic Content Rendering

Products are dynamically displayed using JavaScript by manipulating the DOM. Based on user input (search, filters, sorting), the product list updates in real time, ensuring a responsive and interactive interface.

➤ User Interaction and Feedback

The application provides real-time feedback through UI updates such as cart count badges, toast notifications, and dynamic price calculations. These features improve usability and keep users informed about their actions.

➤ **Cart and Checkout Simulation**

A cart system is implemented where users can add/remove items, adjust quantities, and view total cost. Although no real payment gateway is used, the system simulates the checkout process.

➤ **Data Management using Mock Data**

A predefined dataset of products is used as a mock database. This allows simulation of real-world e-commerce functionality without requiring backend integration.

➤ **Requirement Analysis**

User requirements were identified, such as product browsing, search functionality, filtering options, cart management, and wishlist features. These requirements guided the design and development process to ensure a complete e-commerce experience.

➤ **User Authentication Simulation**

Login and signup functionalities are implemented using local storage to simulate user authentication, allowing personalized interactions within the application.

➤ **Performance Optimization**

Efficient DOM updates and structured code design are used to ensure smooth performance and fast response to user interactions.

➤ **Testing and Validation**

The application was tested for various user interactions, including adding/removing items from the cart, applying filters, searching products, and navigating between pages. Testing ensured smooth functionality, correct data updates, and responsive design across different screen sizes.

➤ **Final Review and Enhancements**

After testing, enhancements such as animations, hover effects, responsive layouts, and improved UI elements were added. These improvements ensure that the application is visually appealing, user-friendly, and provides an engaging shopping experience.

5. Screenshots of code and Result

```
index.html x
index.html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>ShopNova India's Favourite Store by H&R corp.</title>
7   <link href="https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&f=...">
8   <style>
9     :root {
10       --nova-orange: #f97316;
11       --nova-deep: #1e1b4b;
12       --nova-blue: #2563eb;
13       --nova-light-blue: #dbeafe;
14       --nova-yellow: #fbbf24;
15       --nova-green: #16a34a;
16       --nova-red: #dc2626;
17       --bg: #f1f5f9;
18       --card: #ffffff;
19       --text: #0f172a;
20       --muted: #64748b;
21       --border: #e2e8f0;
22       --nav-h: 60px;
23     }
24
25     * {
```

```
index.html x
index.html
2 <html lang="en">
3 <head>
8 <style>
284
285 /* Account Dropdown */
286 .account-wrap {
287   position: relative;
288 }
289
290 .account-dropdown {
291   position: absolute;
292   top: 100%;
293   right: 0;
294   background: white;
295   border-radius: 8px;
296   box-shadow: 0 8px 24px rgba(0, 0, 0, .15);
297   min-width: 240px;
298   margin-top: 12px;
299   display: none;
300   animation: fadeIn .2s;
301 }
302
303 .account-dropdown.show {
304   display: block;
305 }
```

```

2 <html lang="en">
1680 <body>
2003 <script>
2763 //
2763 function toggleAcctDrop(e) {
2764     e.stopPropagation();
2765     document.getElementById('acctDrop').classList.toggle('show');
2766 }
2767
2768 function closeAcctDrop() {
2769     document.getElementById('acctDrop').classList.remove('show');
2770 }
2771
2772 document.addEventListener('click', function(e) {
2773     if (!e.target.closest('.account-wrap')) {
2774         closeAcctDrop();
2775     }
2776 });
2777
2778 // =====
2779 // TOAST
2780 // =====
2781 function showToast(msg, type = 'success') {
2782     const t = document.getElementById('toast');
2783     t.className = 'toast ' + type;
2784     document.getElementById('toast').classList.add('show');

```

```

index.html.html 2 x
index.html.html > html > body > script > openSignup
2 <html lang="en">
1680 <body>
2003 <script>
2737 function updateAccountUI() {
2738     if (currentUser) {
2739         document.getElementById('acctGreeting').textContent = 'Hello, ' + currentUser.name;
2740         document.getElementById('loggedOut').style.display = 'none';
2741         document.getElementById('loggedIn').style.display = 'block';
2742         document.getElementById('acctDispName').textContent = currentUser.name;
2743         document.getElementById('acctDispEmail').textContent = currentUser.email;
2744     } else {
2745         document.getElementById('acctGreeting').textContent = 'Hello, Sign in';
2746         document.getElementById('loggedOut').style.display = 'block';
2747         document.getElementById('loggedIn').style.display = 'none';
2748     }
2749 }
2750
2751 function showAccountDirect() {
2752     if (!currentUser) {
2753         openLogin();
2754         return;
2755     }
2756     closeAcctDrop();
2757     showToast('Account page coming soon!', 'success');
2758 }

```

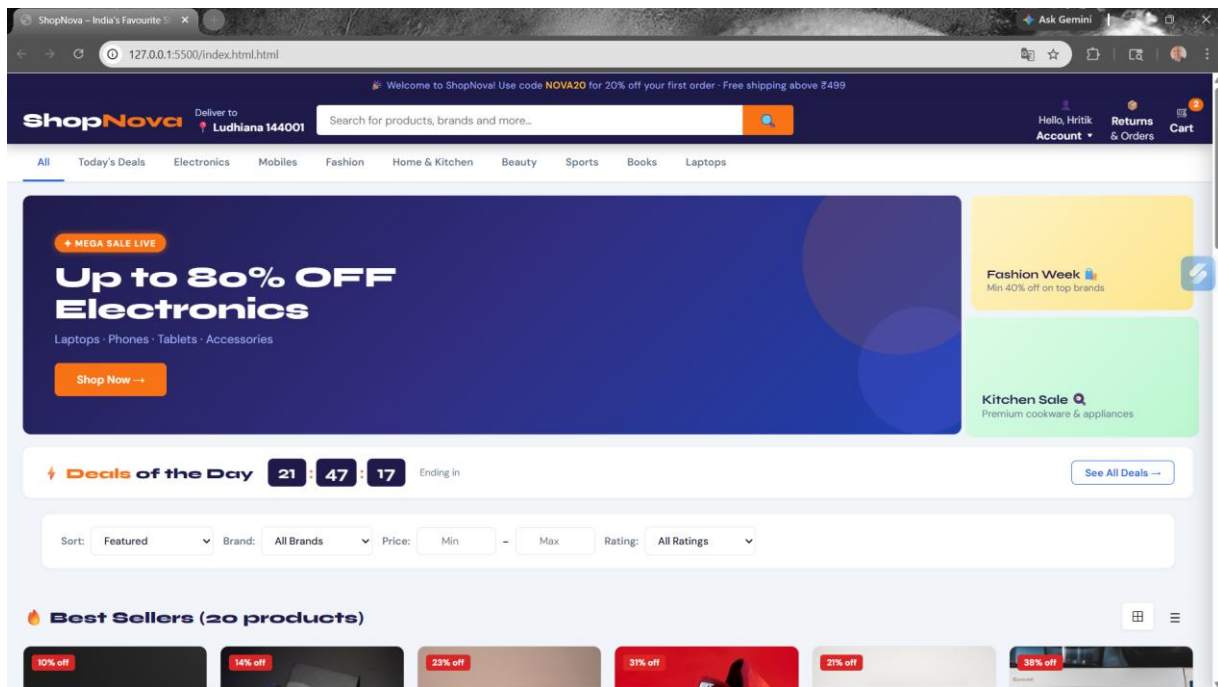


FIG : 1 Front Page

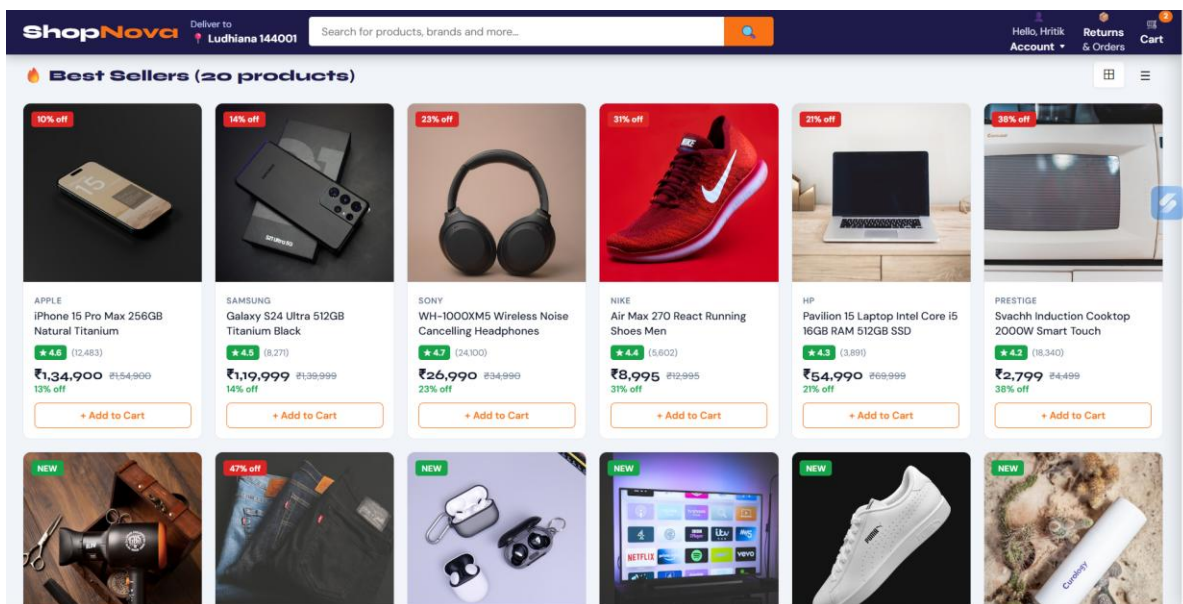


FIG : 2 Products Available

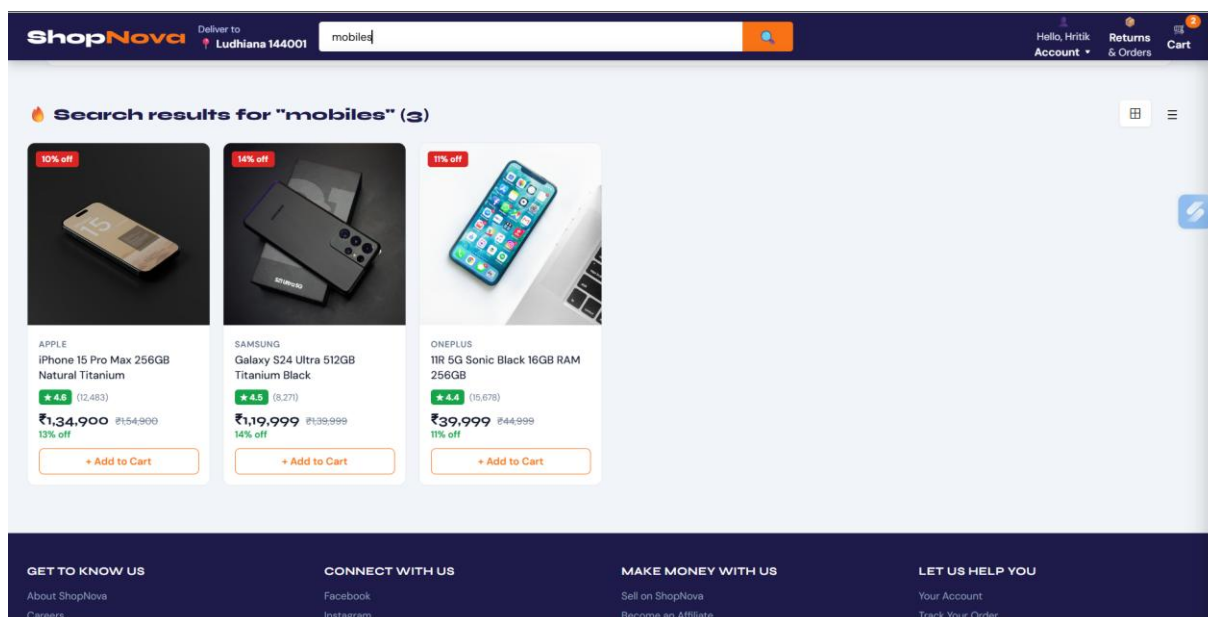


FIG : 3 Search Bar (Mobiles)

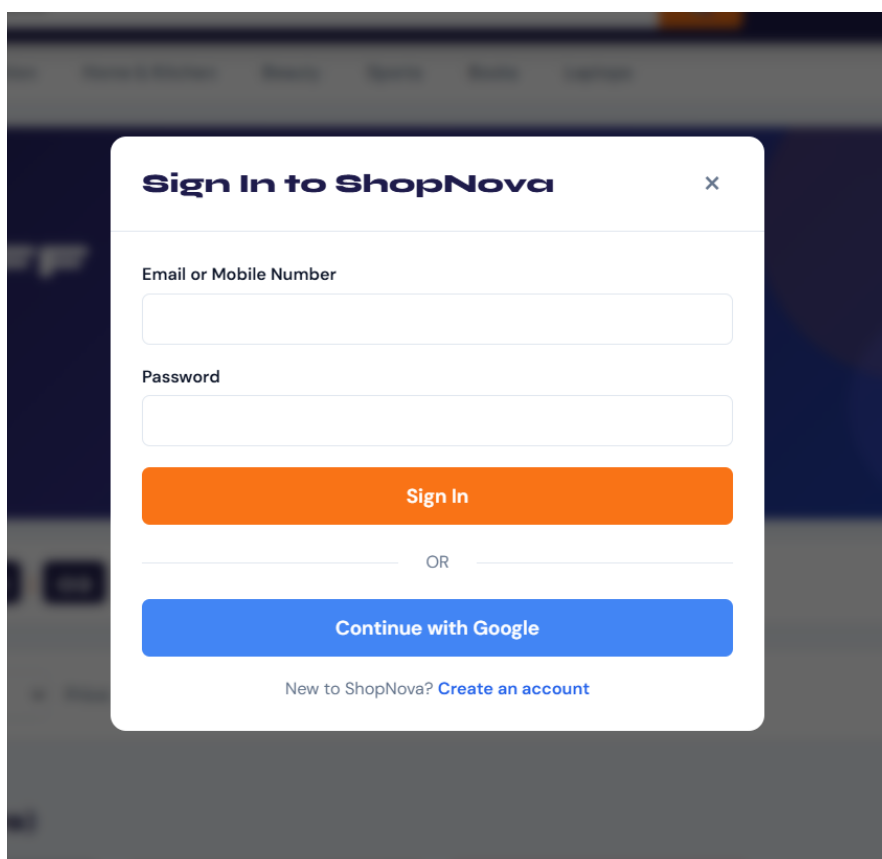
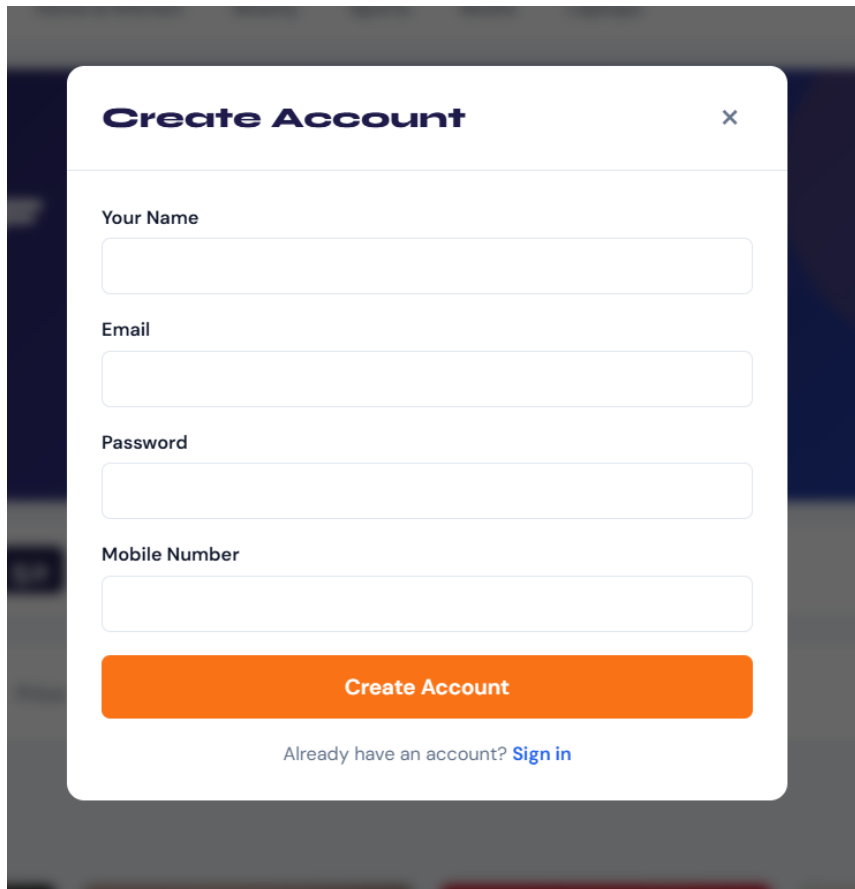


FIG : 4 Sign In



A modal form titled "Create Account" with a close button (X) in the top right corner. The form contains four input fields: "Your Name", "Email", "Password", and "Mobile Number". Below the input fields is a large orange button labeled "Create Account". At the bottom of the modal, there is a link that says "Already have an account? Sign in".

FIG : 5 Create Account

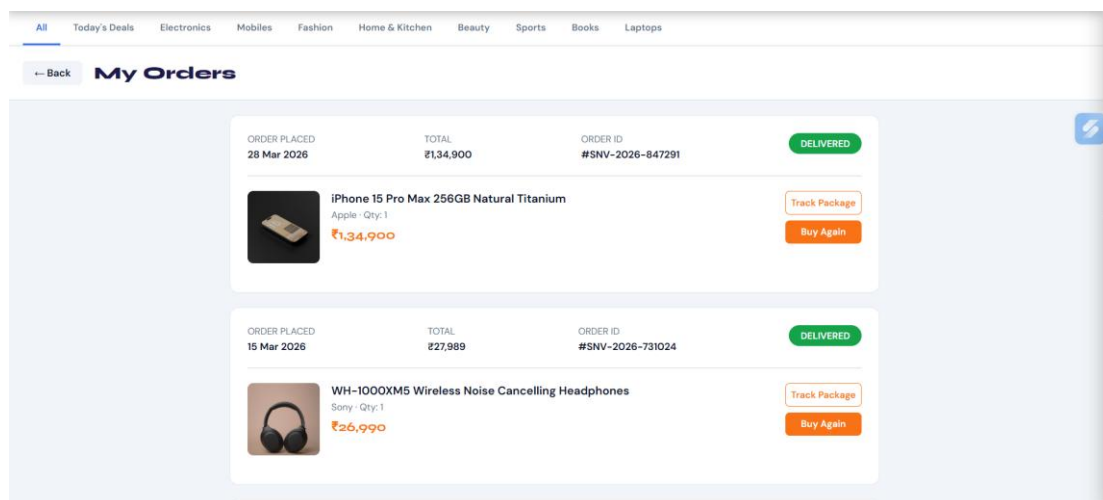


FIG : 6 My Orders

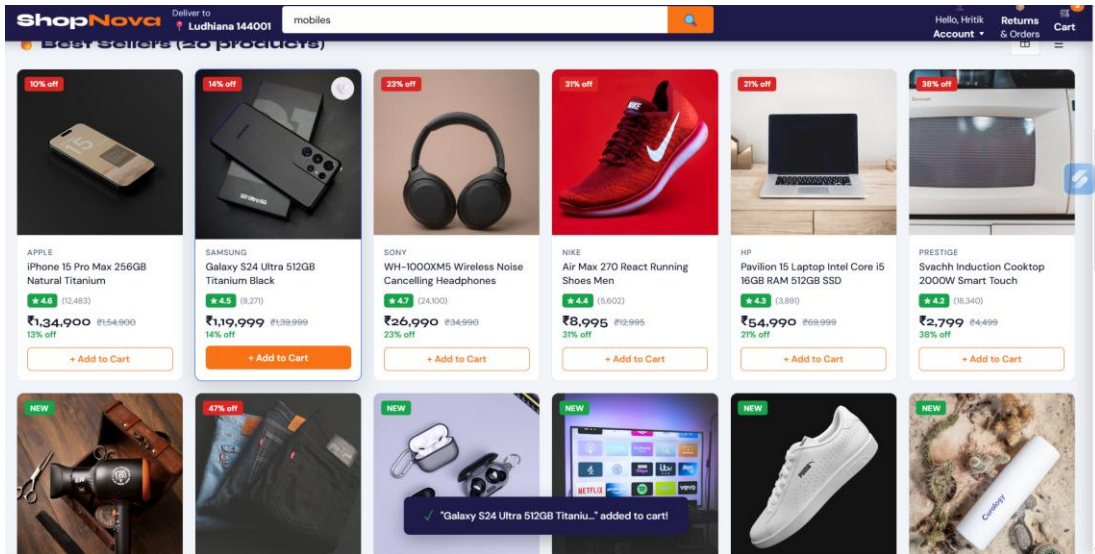


FIG : 7 Items Added To Cart

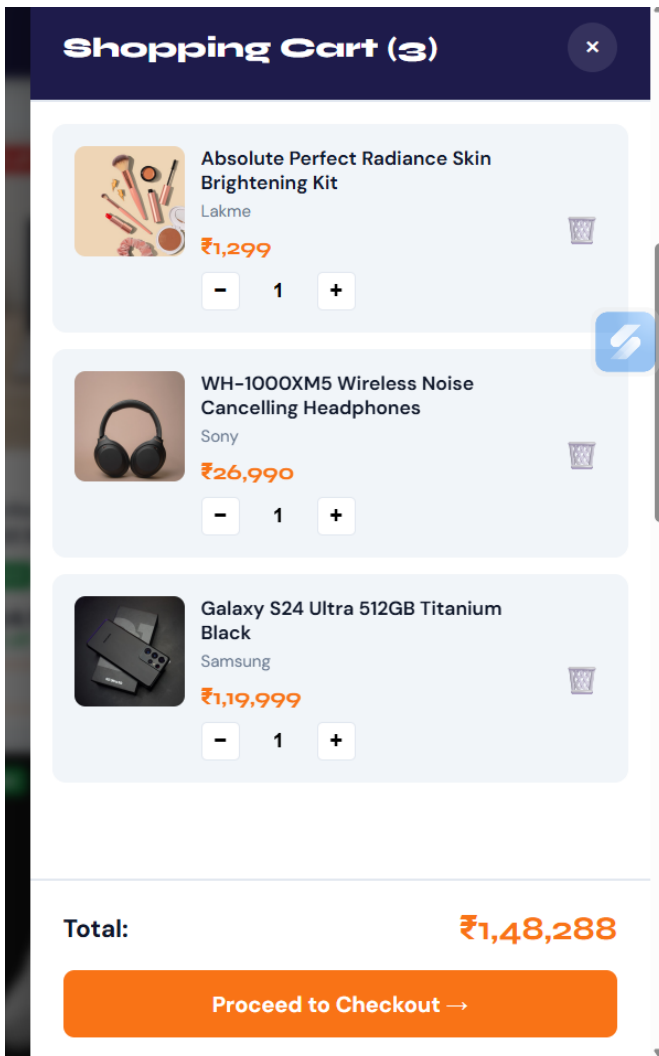


FIG : 8 Items In Cart

5. Conclusion

The project “**ShopNova: E-Commerce Web App**” successfully demonstrates how modern web technologies can be used to develop an interactive, user-friendly, and feature-rich online shopping platform. By utilizing **HTML, CSS, and JavaScript**, the application provides a dynamic interface that allows users to browse products, apply filters, search items, and manage their shopping activities efficiently.

This project showcases how fundamental frontend development concepts such as **DOM manipulation, event handling, conditional logic, and responsive design** can be integrated to create a complete and functional web application. The modular structure of the code ensures clarity, maintainability, and scalability for future enhancements. Features such as real-time cart updates, wishlist management, search suggestions, and filtering mechanisms make the system both practical and engaging.

The application highlights the importance of **user experience (UI/UX)** in web development by providing smooth navigation, interactive elements, and responsive layouts. It demonstrates how client-side technologies can effectively simulate real-world e-commerce functionalities without backend integration.

In the future, this project can be enhanced by adding features such as backend integration, database connectivity, secure authentication systems, payment gateway integration, and advanced functionalities like product recommendations and order tracking.

Overall, the project serves as a strong foundation for understanding e-commerce systems and applying frontend development skills to real-world software applications.

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